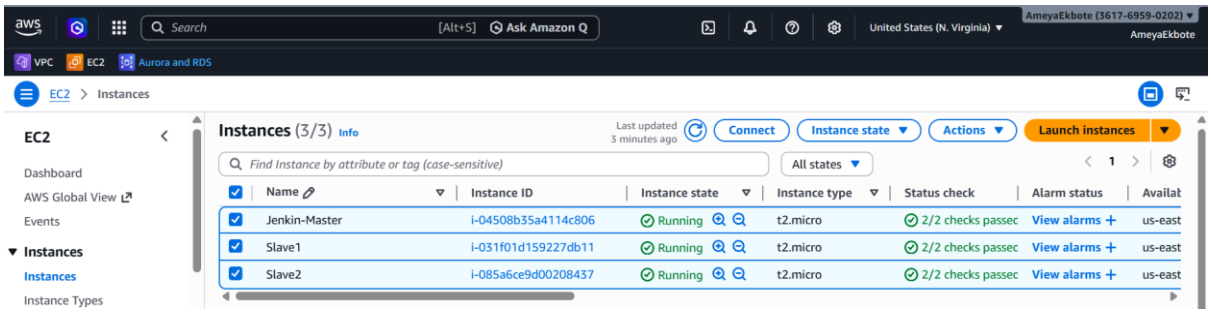
Assignment 1 =>

Tasks To Be Performed:

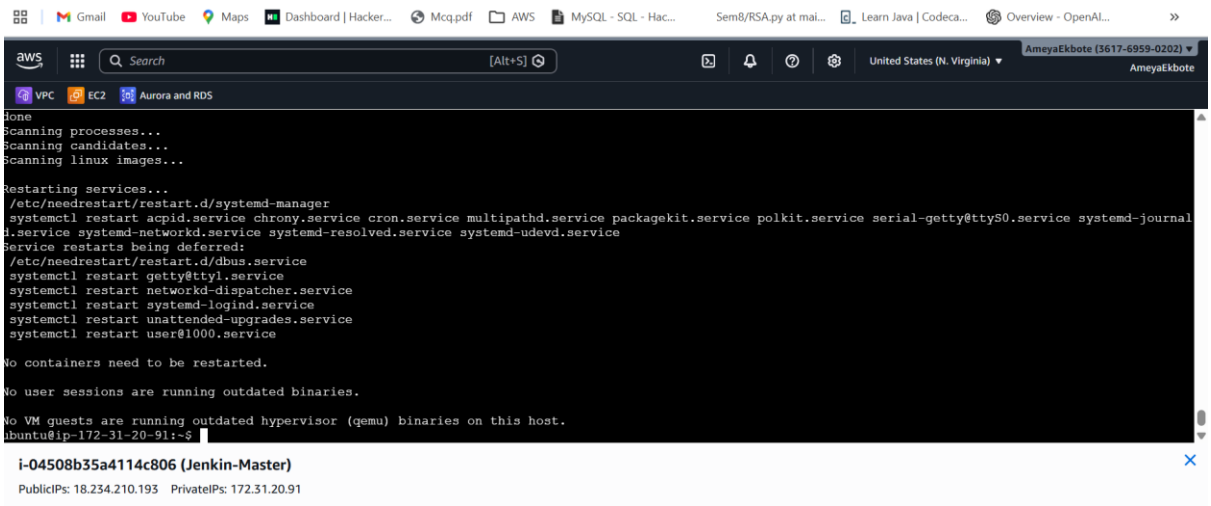
1. Trigger a pipeline using Git when push on develop branch
2. Pipeline should pull Git content to a folder

Installation of jenkins and create nodws , jobs =>

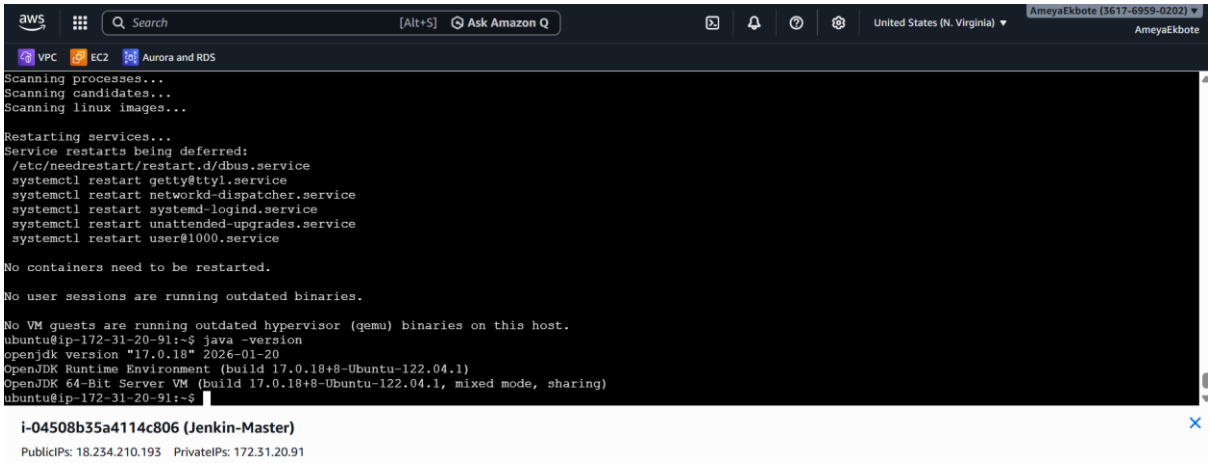
Created 3 instances =>




```
sudo apt update //all 3 instances
```



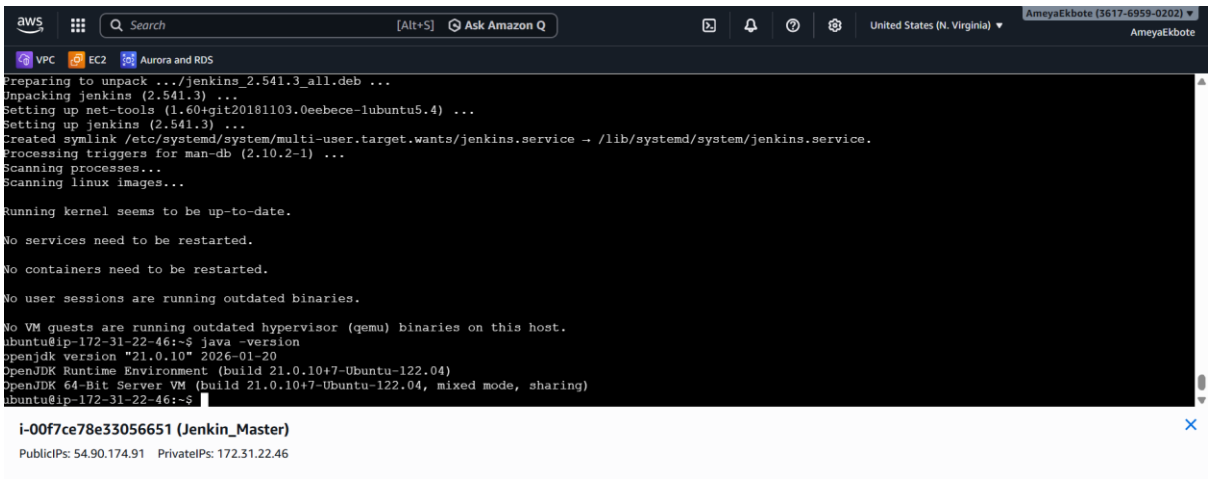

```
sudo apt-get install fontconfig openjdk-21-jre // all 3 instances
```

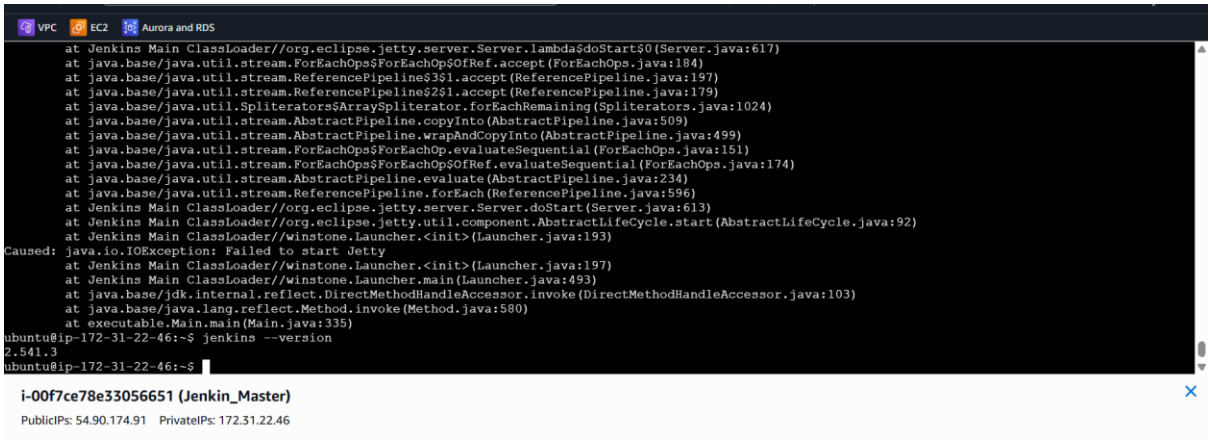


only on master node >

https://pkg.jenkins.io/debian-stable/

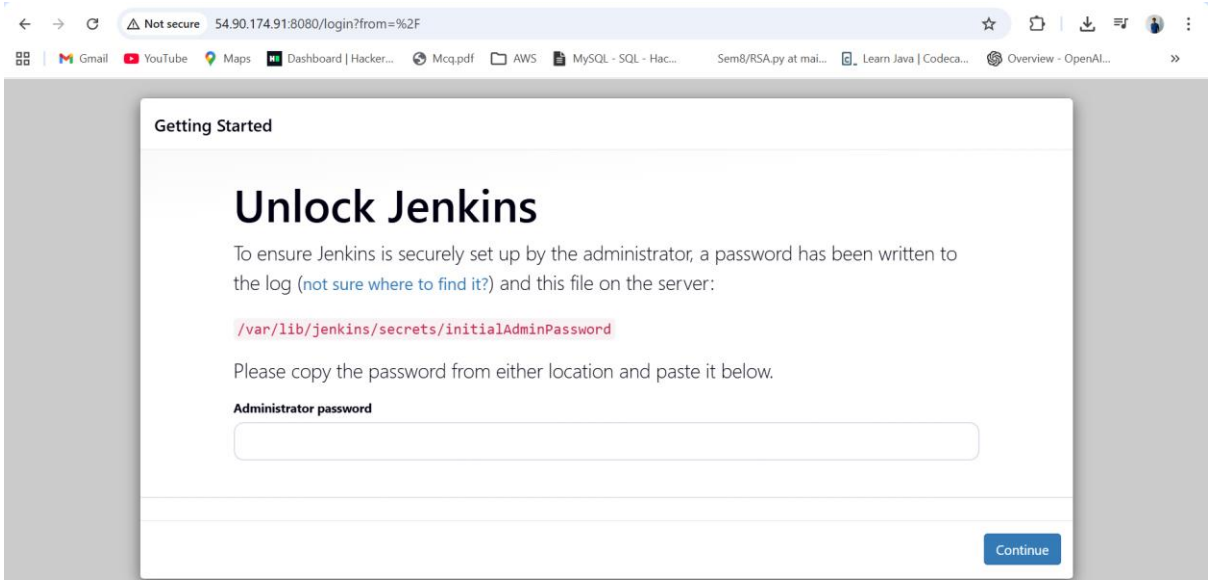
```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
    https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key


echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
    https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
    /etc/apt/sources.list.d/jenkins.list > /dev/null


sudo apt-get update
  sudo apt-get install fontconfig openjdk-21-jre
  sudo apt-get install jenkins
```





```
VPC  EC2  Aurora and RDS
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.lambda$doStart$0(Server.java:617)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:184)
        at java.base/java.util.stream.ReferencePipeline$3$1.accept(ReferencePipeline.java:197)
        at java.base/java.util.stream.ReferencePipeline$2$1.accept(ReferencePipeline.java:179)
        at java.base/java.util.Spliterators$ArraySpliterator.forEachRemaining(Spliterators.java:1024)
        at java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:509)
        at java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:499)
        at java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:151)
        at java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:174)
        at java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:234)
        at java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:596)
        at Jenkins Main ClassLoader//org.eclipse.jetty.server.Server.doStart(Server.java:613)
        at Jenkins Main ClassLoader//org.eclipse.jetty.util.component.AbstractLifeCycle.start(AbstractLifeCycle.java:92)
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:193)
Caused: java.io.IOException: Failed to start Jetty
        at Jenkins Main ClassLoader//winstone.Launcher.<init>(Launcher.java:197)
        at Jenkins Main ClassLoader//winstone.Launcher.main(Launcher.java:493)
        at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(DirectMethodHandleAccessor.java:103)
        at java.base/java.lang.reflect.Method.invoke(Method.java:580)
        at executable.Main.main(Main.java:335)
ubuntu@ip-172-31-22-46:~$ jenkins --version
2.541.3
ubuntu@ip-172-31-22-46:~$
i-00f7ce78e33056651 (Jenkin_Master)
PublicIPs: 54.90.174.91  PrivateIPs: 172.31.22.46
```

http://54.90.174.91:8080/



sudo cat /var/lib/jenkins/secrets/initialAdminPassword

Not secure 54.90.174.91:8080

Gmail YouTube Maps Dashboard | Hacker... Mcq.pdf AWS MySQL - SQL - Ha

Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

Start using Jenkins

Not secure 54.90.174.91:8080/manage/configureSecurity/

Gmail YouTube Maps Dashboard | Hacker... Mcq.pdf AWS MySQL - SQL - Hac...

Jenkins / Manage Jenkins / Security

Shows descriptions mostly as written. HTML unsafe characters like < and & are escaped to the converted to their HTML equivalent.

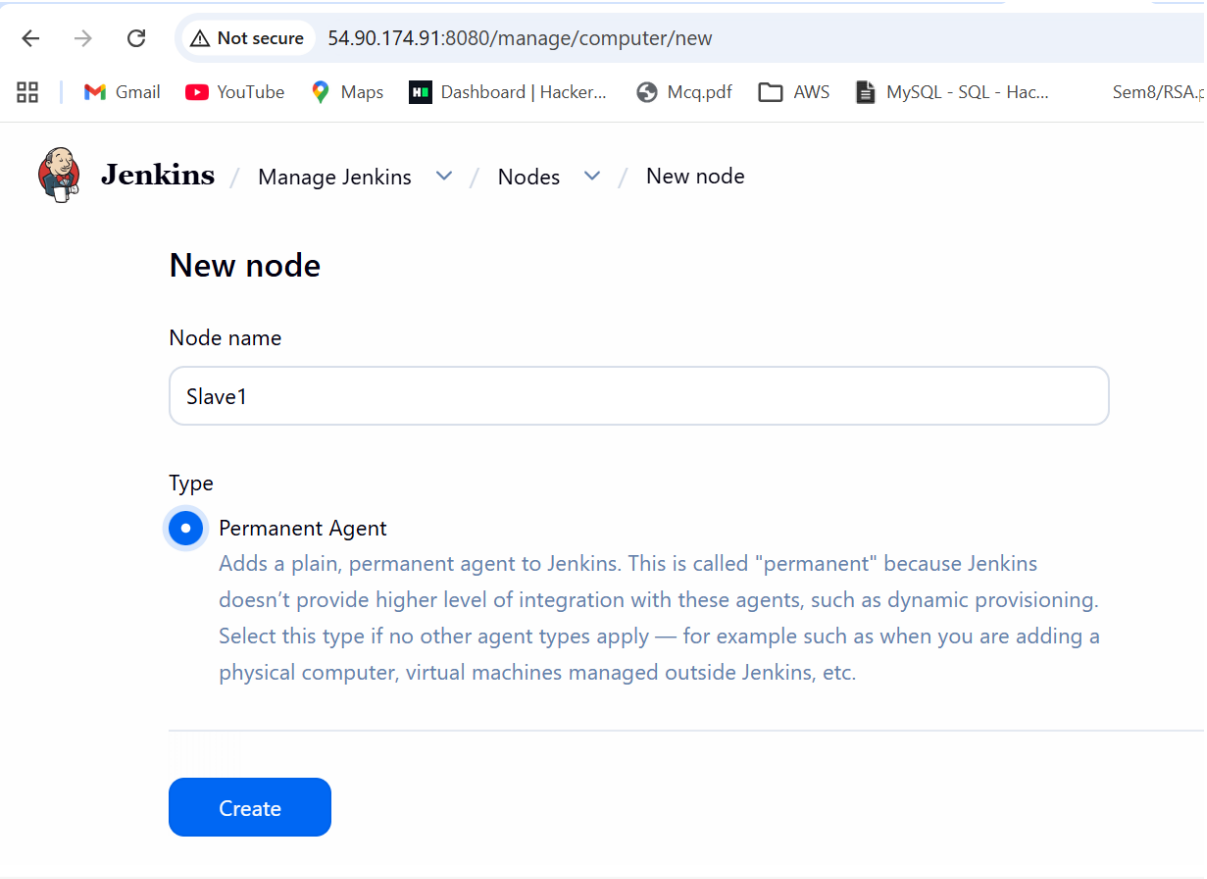
Agents

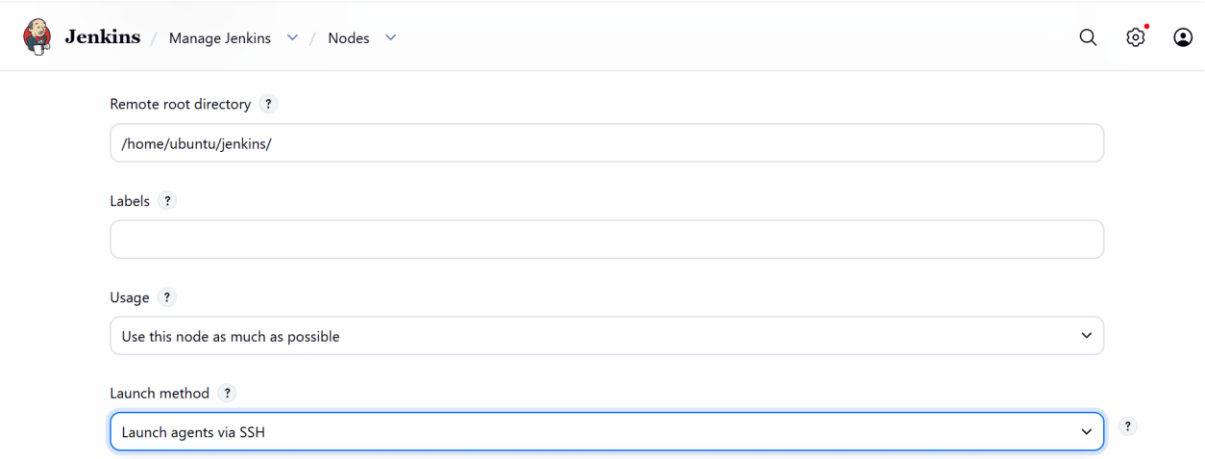
TCP port for inbound agents ?

- Fixed
- Random
- Disable

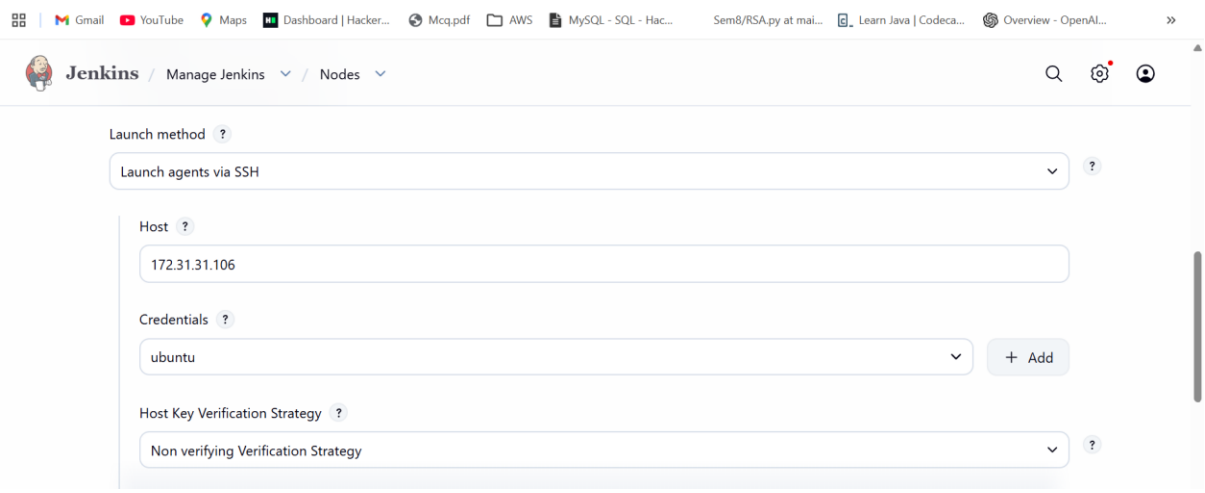
CSRF Protection

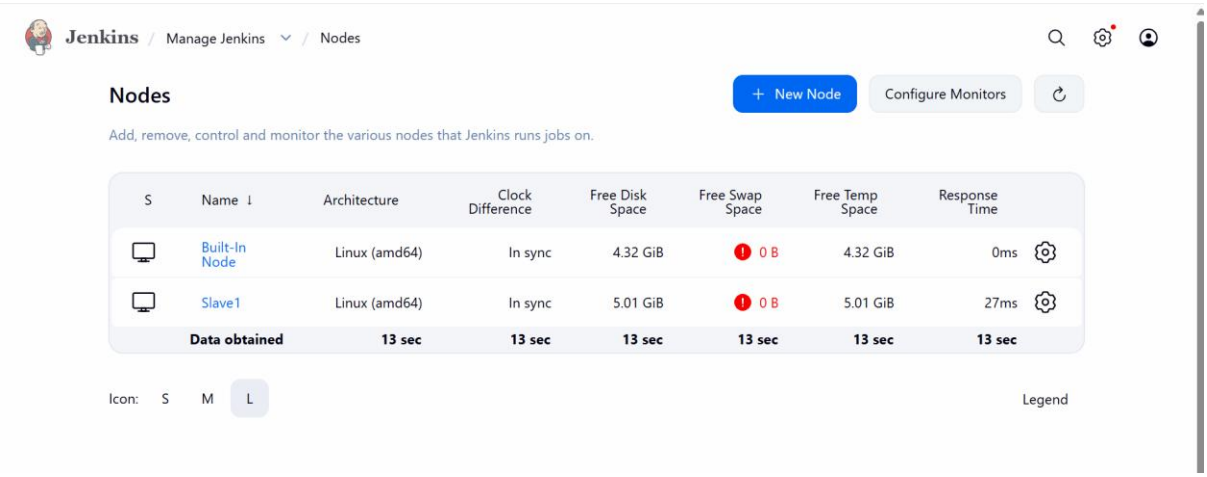
Save Apply



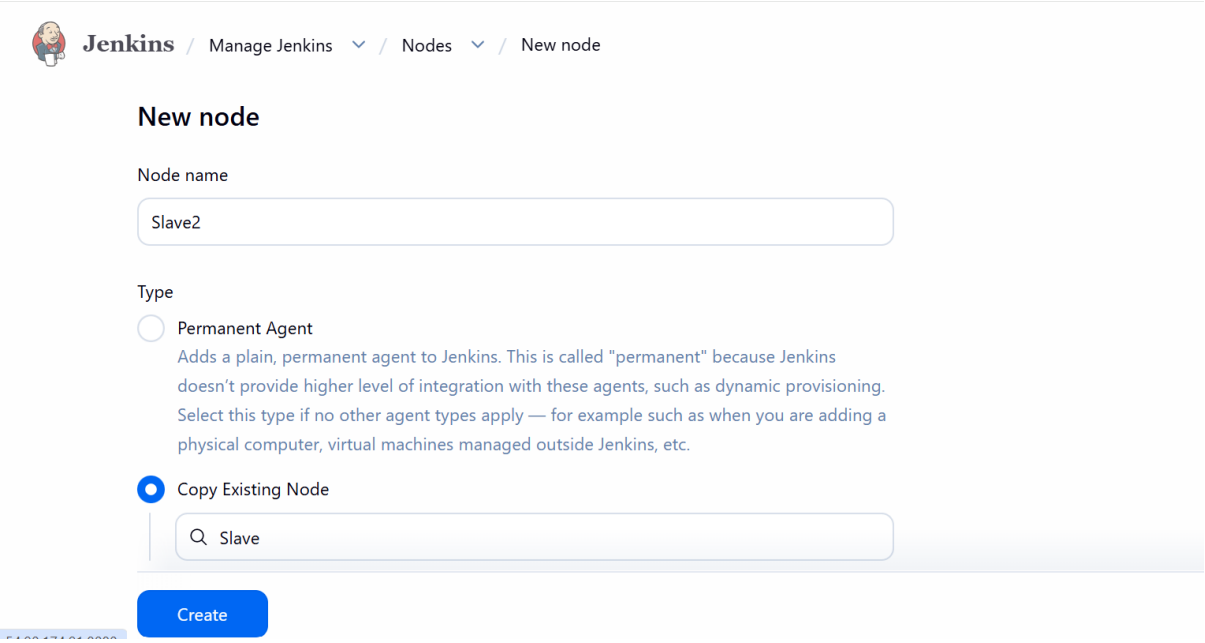


Add privte IP of slave1 node and .pem file(used in instance creation) private key create with username as ubuntu

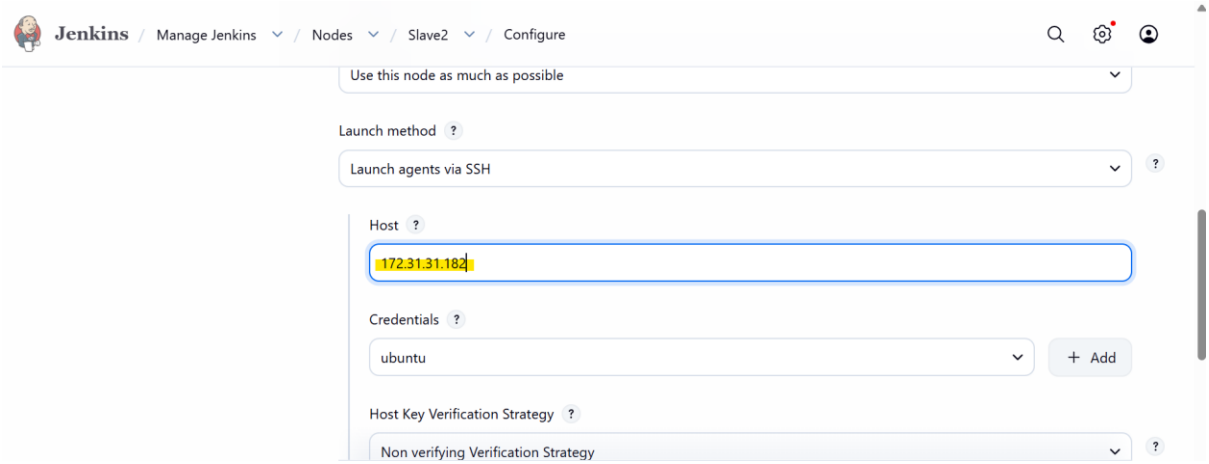


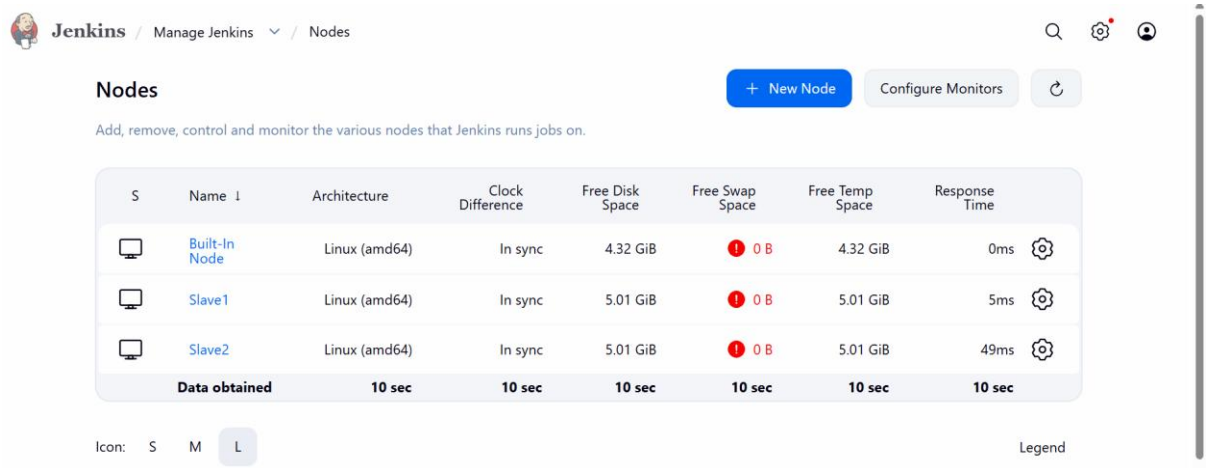


Slave 2 =>

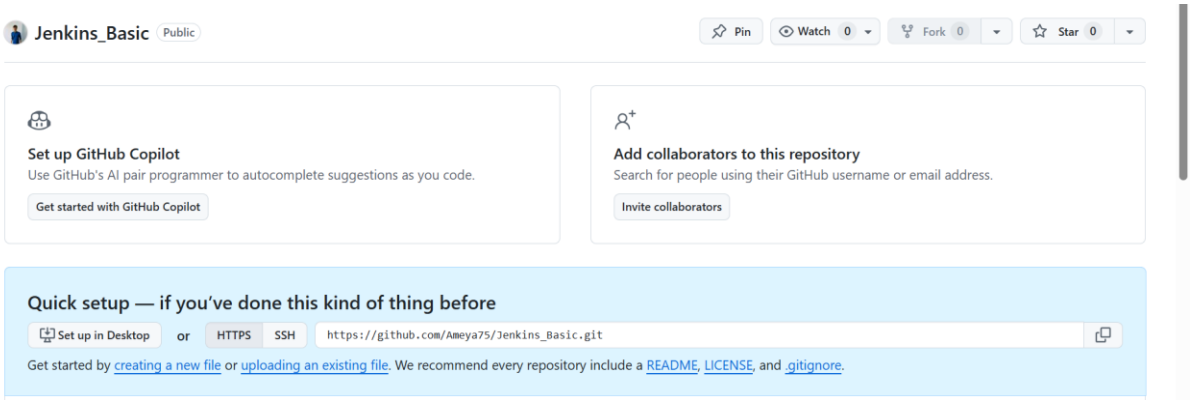


Just add private IP of Slave2 instance in host





Create JOB => Job1 => not selecting pipeline for 1st assignment* .



In Jenkins GUI add this URL in JOB =>

Jenkins / Job1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Project url ?

https://github.com/Ameya75/Jenkins_Basic.git

Advanced

This project is parameterized ?

Throttle builds ?

Execute concurrent builds if necessary ?

Restrict where this project can be run ?

Label Expression ?

Slave1

Save Apply

Jenkins / Job1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

None

Git ?

Repositories ?

Repository URL ?

https://github.com/Ameya75/Jenkins_Basic.git

Credentials ?

- none - + Add

Jenkins / Job1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

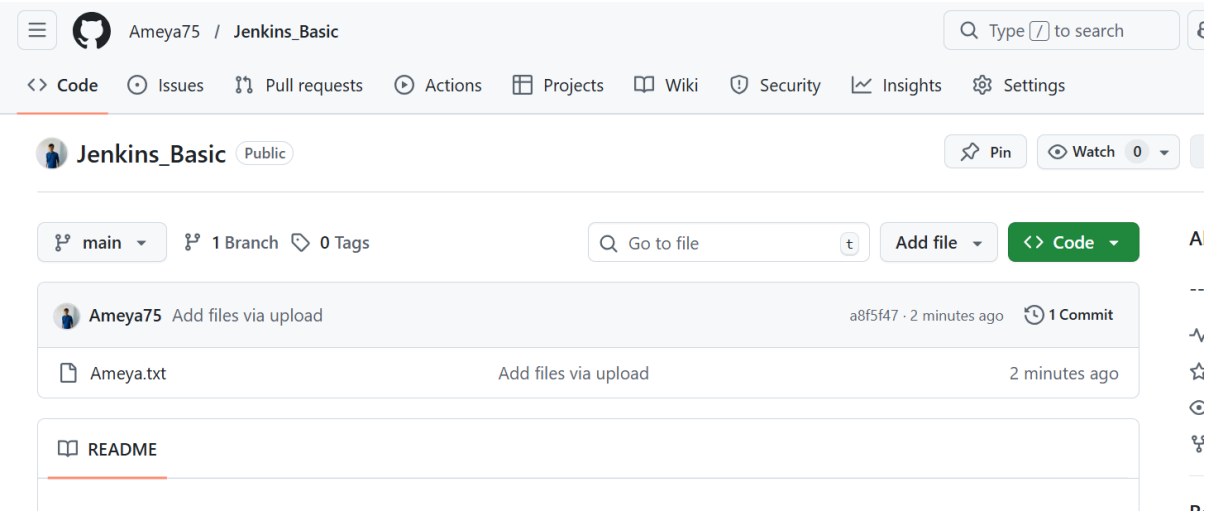
Branches to build ?

Branch Specifier (blank for 'any') ?

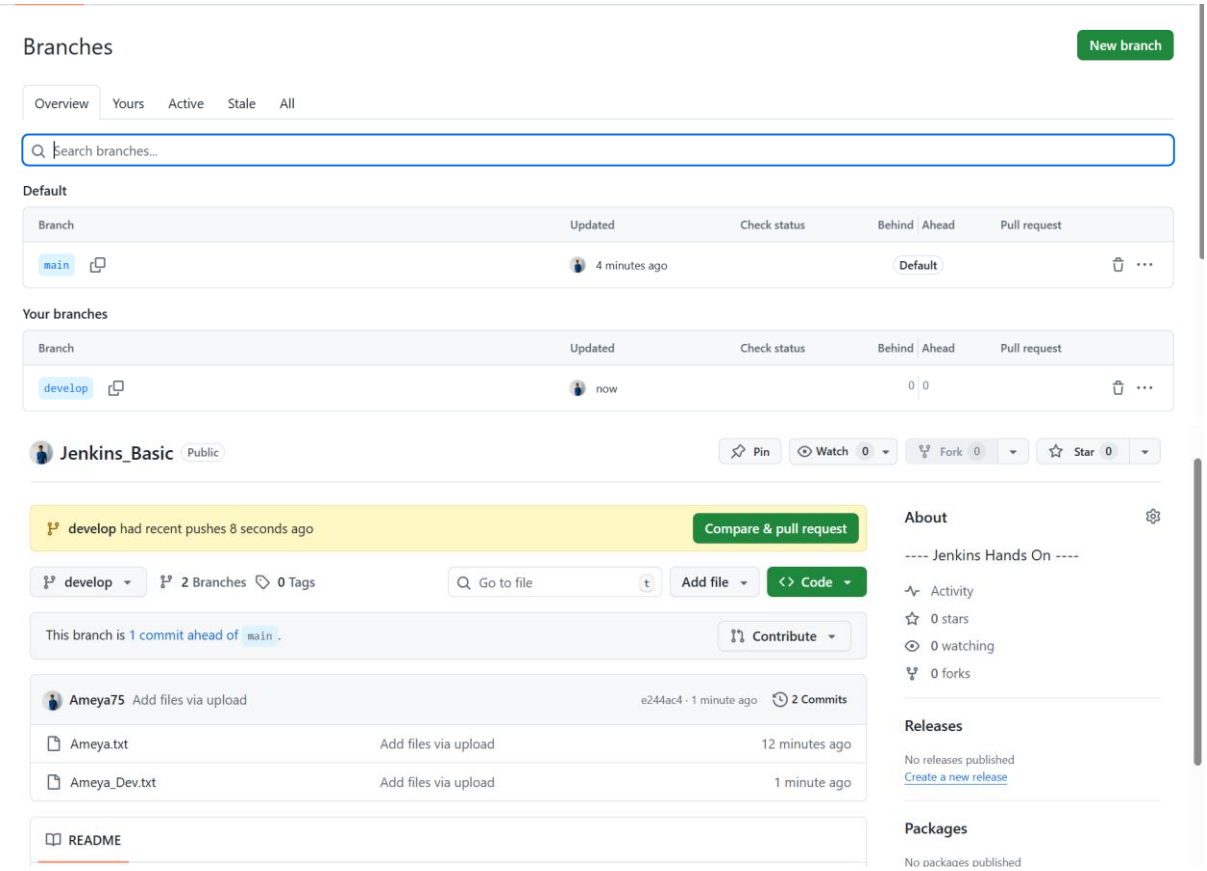
*/develop

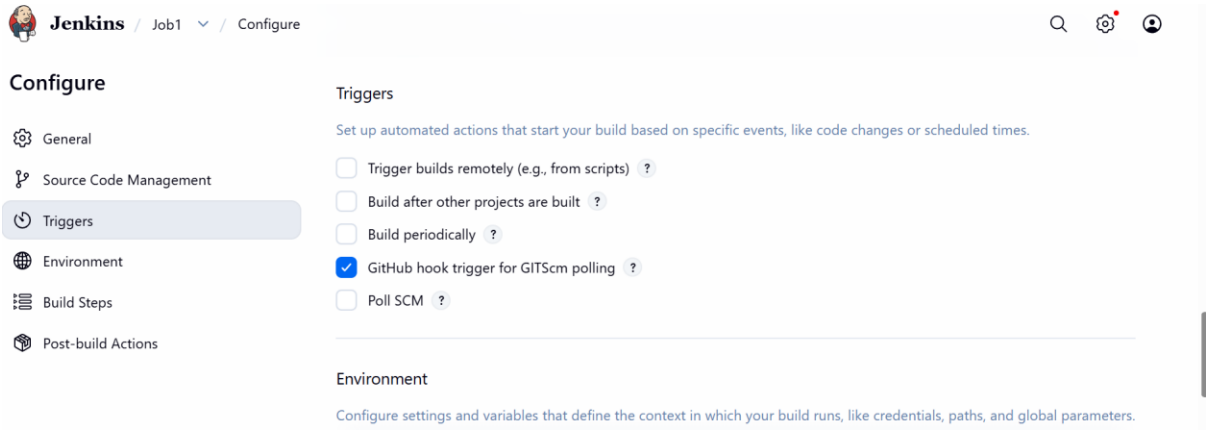
+ Add Branch

Repository browser ?

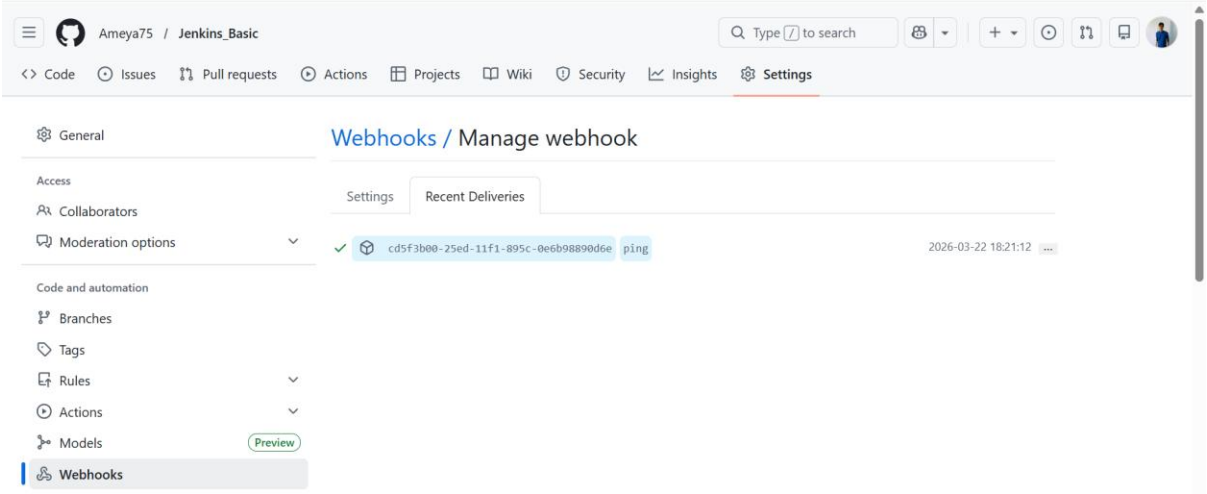


created new branch =>

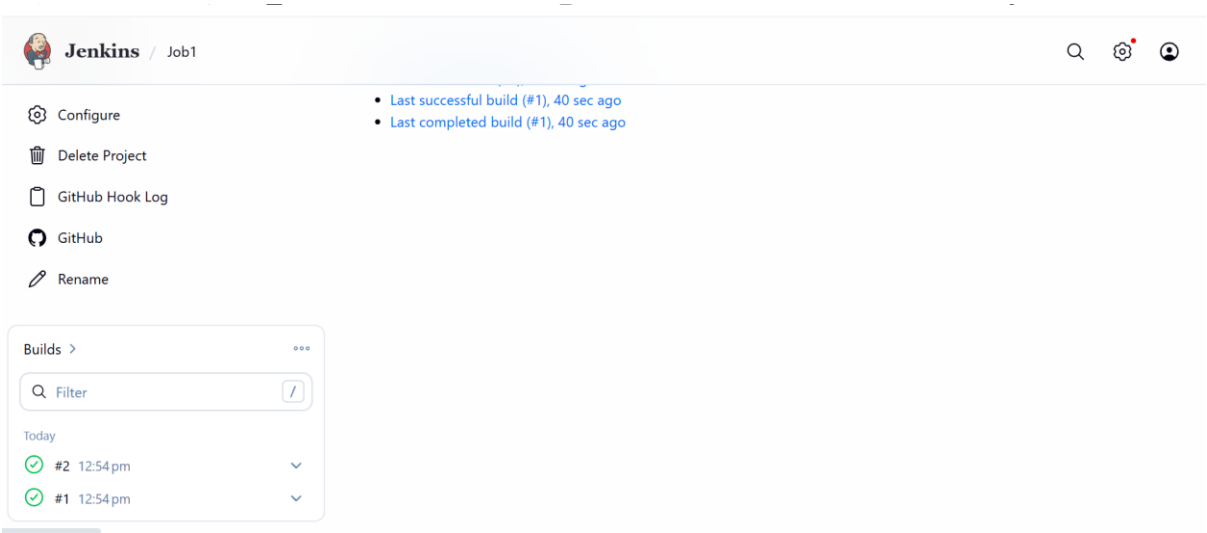




JOB created on jenkins .

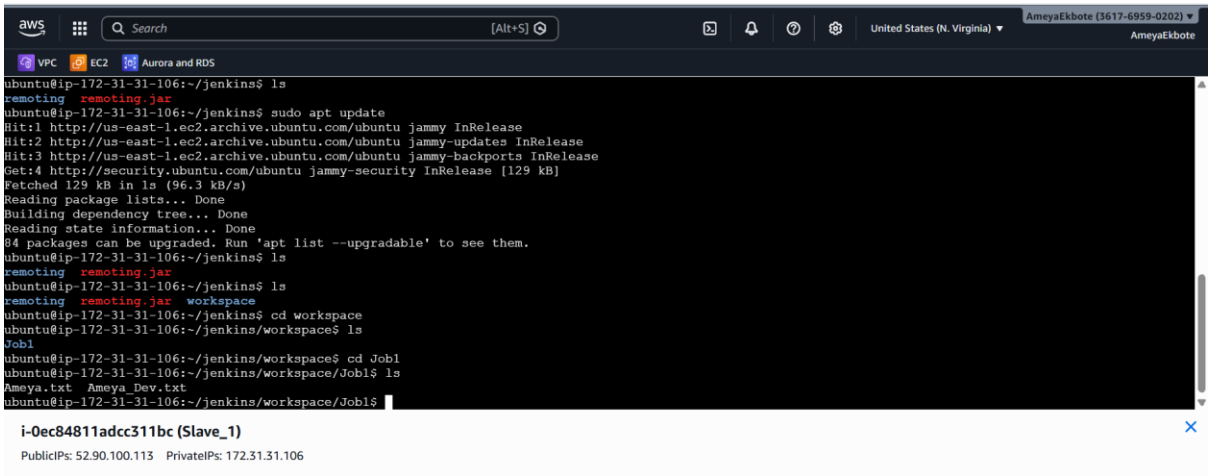


Building JOB =>



Automating on Slave1 for Job1 =>

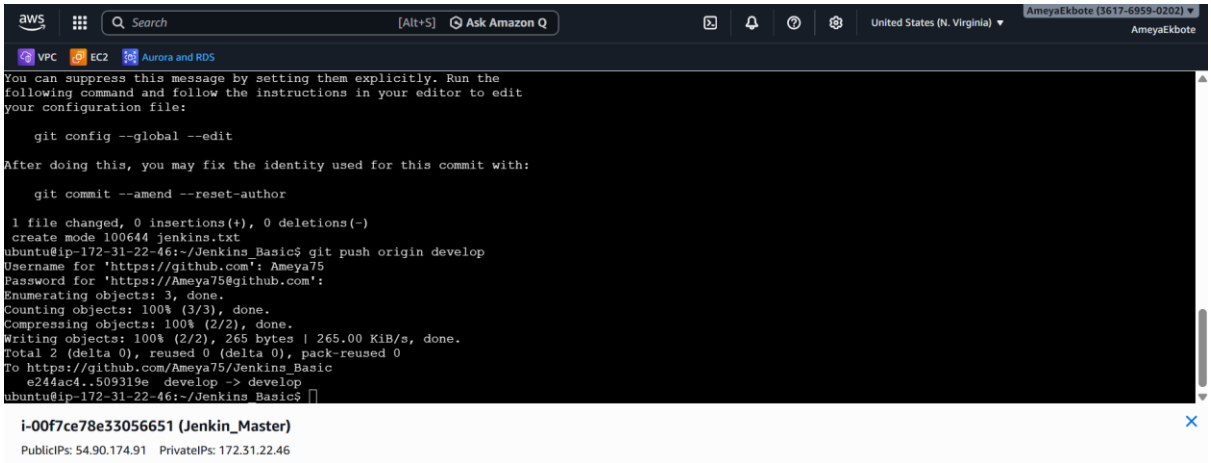
```
ubuntu@ip-172-31-31-106:~$ ls
jenkins
ubuntu@ip-172-31-31-106:~$ cd jenkins
ubuntu@ip-172-31-31-106:~/jenkins$ ls
remoting  remoting.jar  workspace
ubuntu@ip-172-31-31-106:~/jenkins$ cd workspace
ubuntu@ip-172-31-31-106:~/jenkins/workspace$ ls
Job1
ubuntu@ip-172-31-31-106:~/jenkins/workspace$ cd Job1
ubuntu@ip-172-31-31-106:~/jenkins/workspace/Job1$ ls
Ameya.txt  Ameya_Dev.txt
ubuntu@ip-172-31-31-106:~/jenkins/workspace/Job1$
```



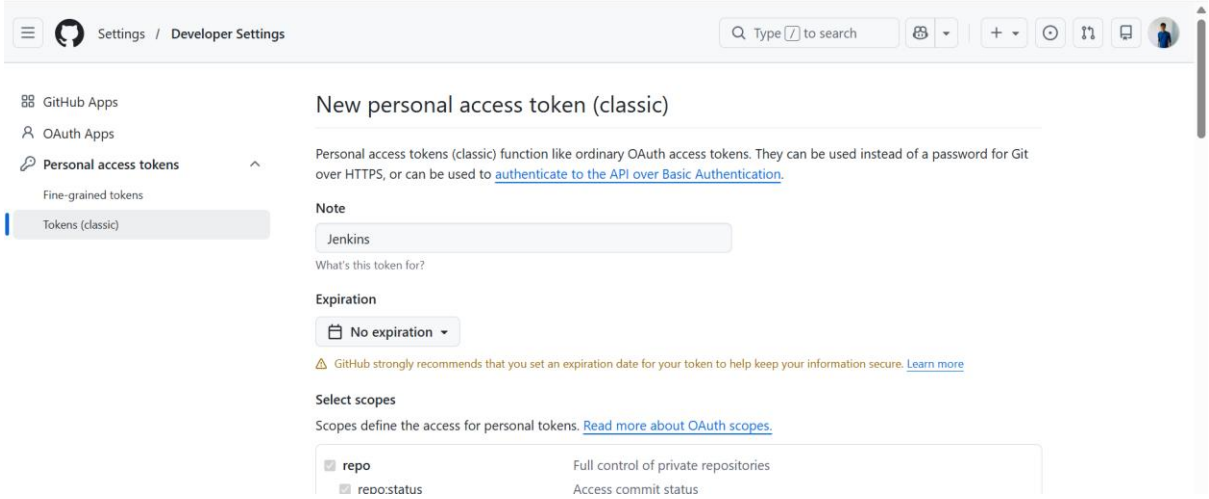
Master node =>

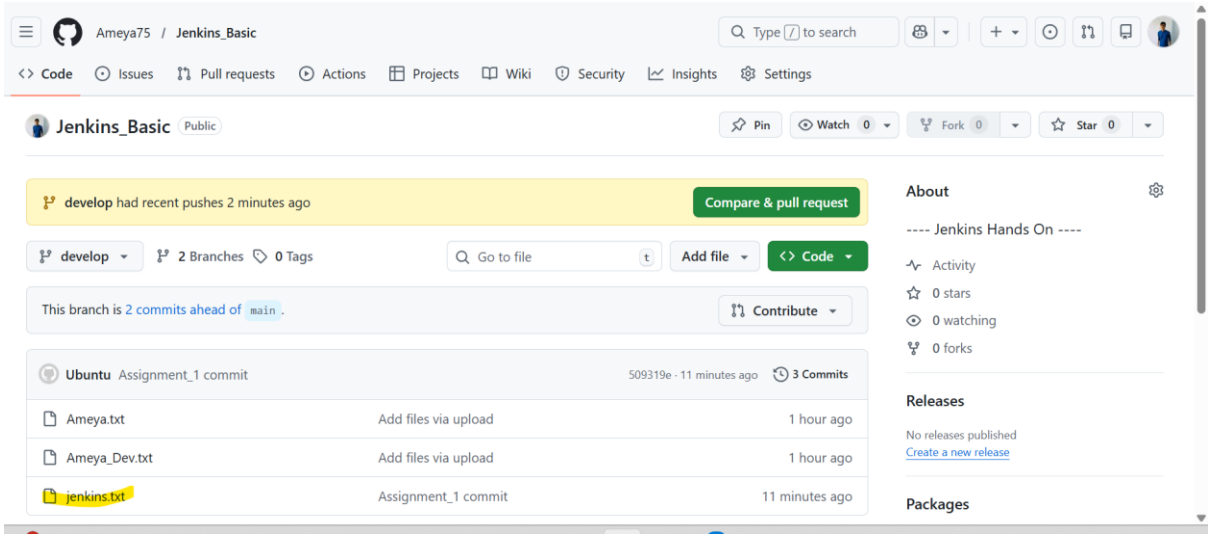
```
ubuntu@ip-172-31-22-46:~$ jenkins --version
2.541.3
ubuntu@ip-172-31-22-46:~$ git clone
https://github.com/Ameya75/Jenkins_Basic
Cloning into 'Jenkins_Basic'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
Receiving objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
ubuntu@ip-172-31-22-46:~$ ls
Jenkins_Basic
ubuntu@ip-172-31-22-46:~$ cd Jenkins_Basic
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
```

```
Ameya.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git branch
* main
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git checkout develop
Branch 'develop' set up to track remote branch 'develop' from 'origin'.
Switched to a new branch 'develop'
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
Ameya.txt  Ameya_Dev.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git branch
* develop
  main
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ touch jenkins.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git add .
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git commit -m"Assignment_1 commit"
[develop 509319e] Assignment_1 commit
 Committer: Ubuntu <ubuntu@ip-172-31-22-46.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 jenkins.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git push origin develop
Username for 'https://github.com': Ameya75
Password for 'https://Ameya75@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 265 bytes | 265.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ameya75/Jenkins_Basic
   e244ac4..509319e  develop -> develop
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$
```

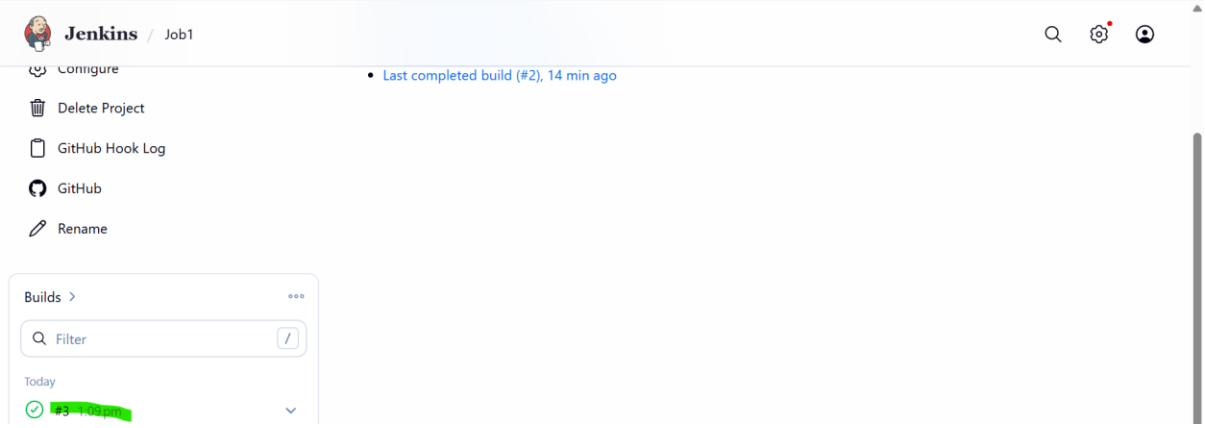


generating token instead of pass on github =>





Automatic trigger verified on jenkins GUI =>



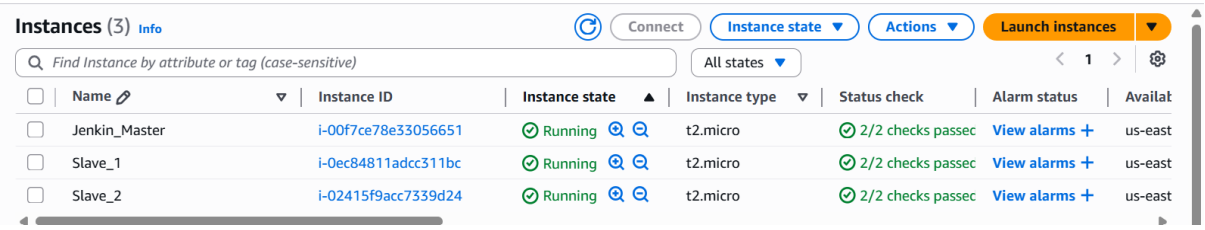
----------END----------

Assignment 2 =>

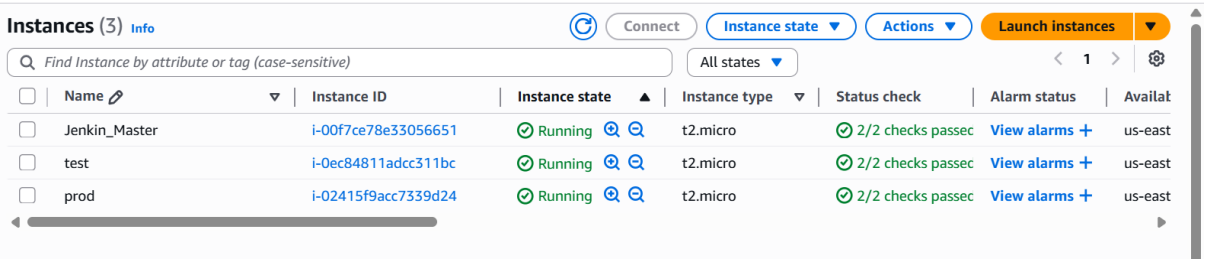
Tasks To Be Performed:

1. Add 2 nodes to Jenkins master
2. Create 2 jobs with the following jobs: a. Push to test b. Push to prod
3. Once a push is made to test branch, copy Git files to test server 4. Once a push is made to master branch, copy Git

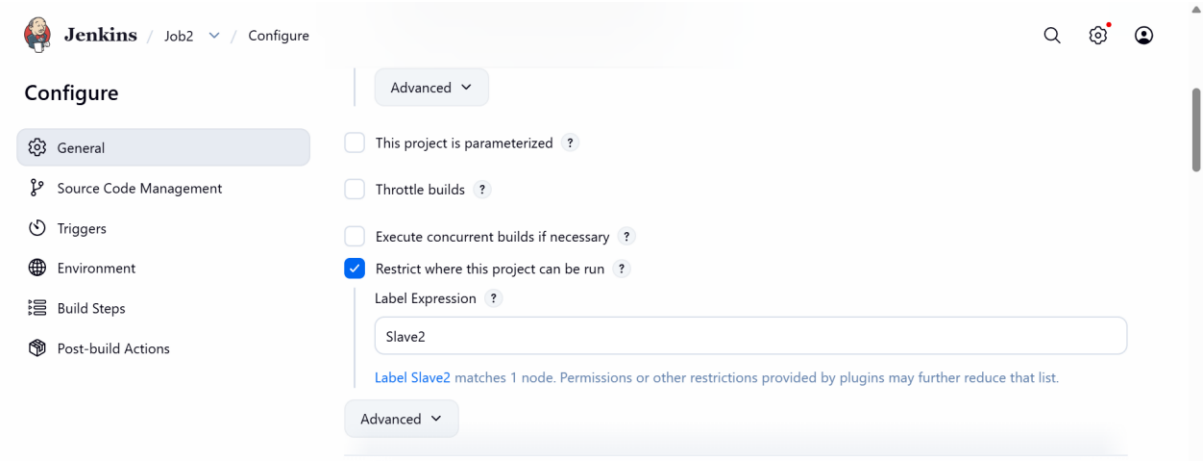
2 nodes already created with name Slave 1 and Slave 2 .



renamed =>



created JOB 2 =>

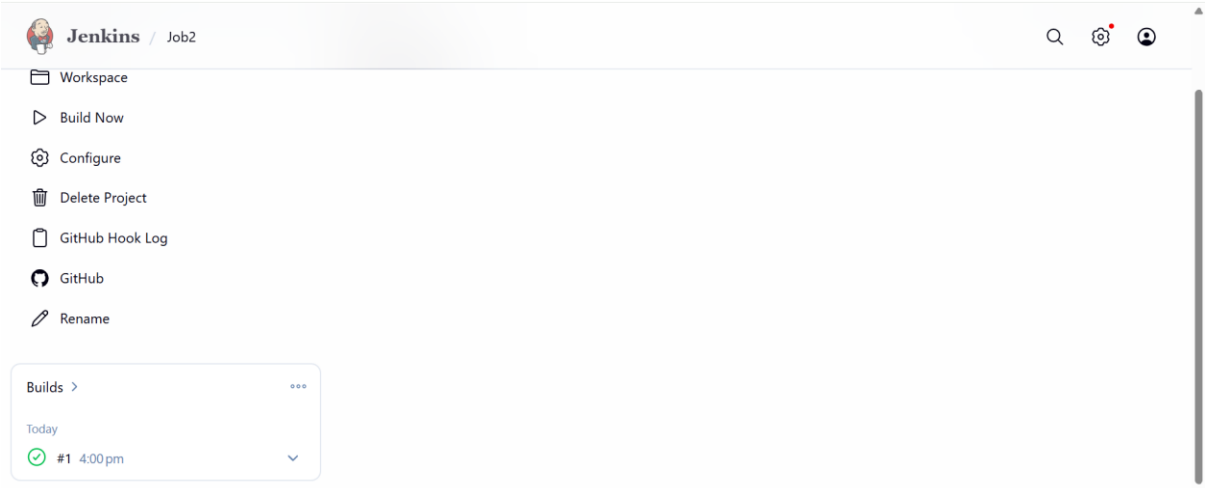


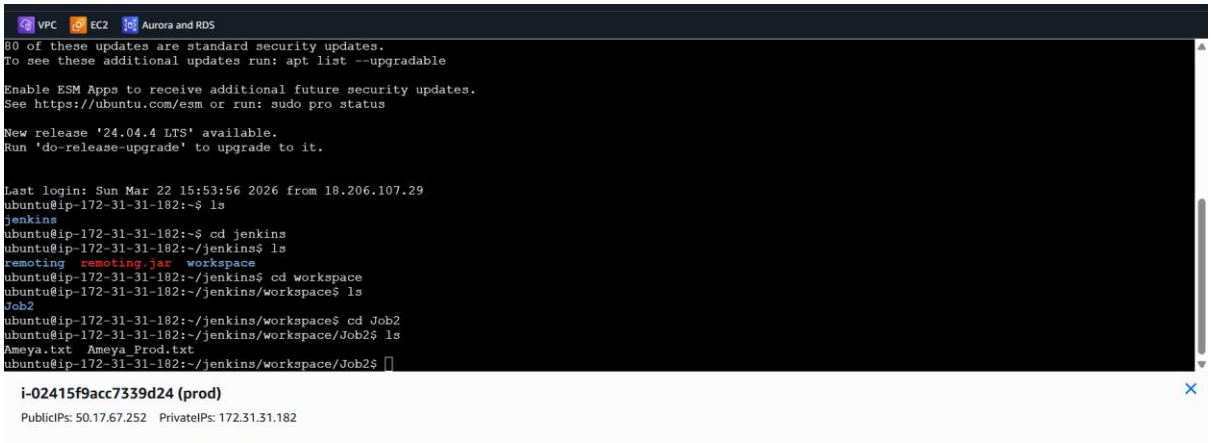
Job 1: Push to Test

- Trigger: GitHub webhook on **test branch**.
- Agent: test node.
- Action: Copy repo files to test server.

  Already completed on Assignment 1 .

- **Job 2: Push to Prod**
- Trigger: GitHub webhook on **main branch**.
- Agent: prod node.
- Action: Copy repo files to prod server





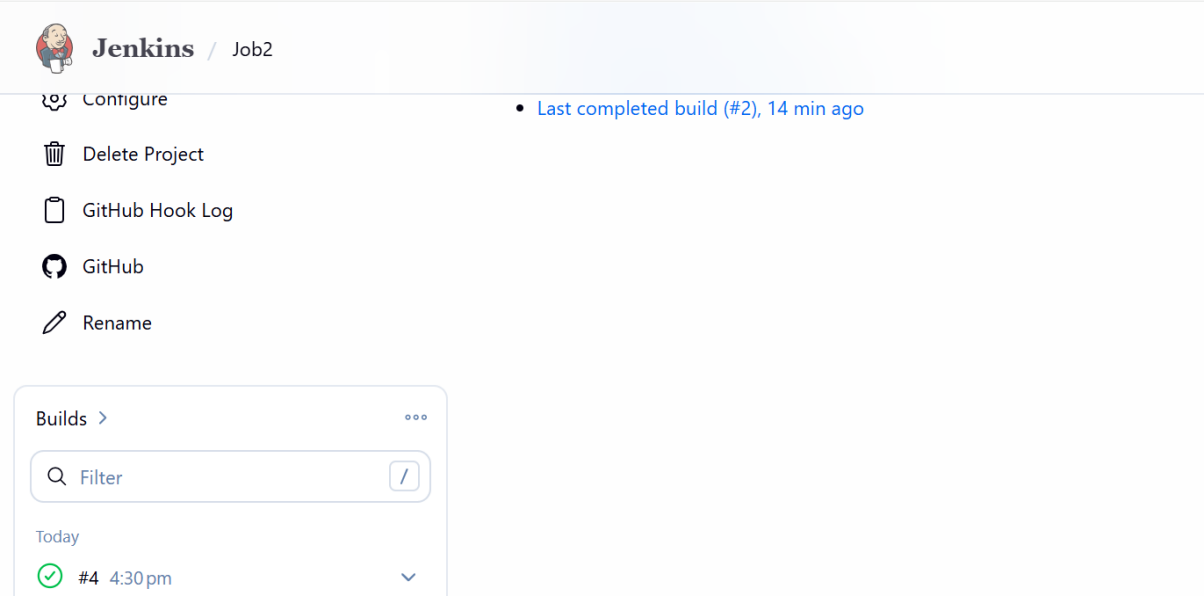
on master creating file to copy

ubuntu@ip-172-31-22-46:~$ ls

- Jenkins_Basic
- ubuntu@ip-172-31-22-46:~$ cd Jenkins_Basic/
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
- Ameya.txt
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git branch
- develop
- * main
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
- Ameya.txt
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git checkout master
- error: pathspec 'master' did not match any file(s) known to git
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ^C
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
- Ameya.txt
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ touch jenkins_prod.txt
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git add .
- ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git commit -m"Assignment_2 commit"
- [main 6b258cf] Assignment_2 commit
- Committer: Ubuntu <ubuntu@ip-172-31-22-46.ec2.internal>
- Your name and email address were configured automatically based
- on your username and hostname. Please check that they are accurate.
- You can suppress this message by setting them explicitly. Run the
- following command and follow the instructions in your editor to edit
- your configuration file:
-
- git config --global --edit
-
- After doing this, you may fix the identity used for this commit with:
-
- git commit --amend --reset-author

```
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 jenkins_prod.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
Ameya.txt  jenkins_prod.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ git push origin main
Username for 'https://github.com': Ameya75
Password for 'https://Ameya75@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 265 bytes | 265.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ameya75/Jenkins_Basic
   a8f5f47..6b258cf  main -> main
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$ ls
Ameya.txt  jenkins_prod.txt
ubuntu@ip-172-31-22-46:~/Jenkins_Basic$
```

Auto trigged =>



On prod aka Slave 2 able to see txt file which was copied from master to prod =>

```
aws   Search [Alt+S]
VPC   EC2   Aurora and RDS
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Sun Mar 22 15:53:56 2026 from 18.206.107.29
ubuntu@ip-172-31-31-182:~$ ls
jenkins
ubuntu@ip-172-31-31-182:~$ cd jenkins
ubuntu@ip-172-31-31-182:~/jenkins$ ls
remoting  remoting.jar  workspace
ubuntu@ip-172-31-31-182:~/jenkins$ cd workspace
ubuntu@ip-172-31-31-182:~/jenkins/workspace$ ls
Job2
ubuntu@ip-172-31-31-182:~/jenkins/workspace$ cd Job2
ubuntu@ip-172-31-31-182:~/jenkins/workspace/Job2$ ls
Ameya.txt  Ameya_Prod.txt
ubuntu@ip-172-31-31-182:~/jenkins/workspace/Job2$ ls
Ameya.txt
ubuntu@ip-172-31-31-182:~/jenkins/workspace/Job2$ ls
Ameya.txt
ubuntu@ip-172-31-31-182:~/jenkins/workspace/Job2$ ls
Ameya.txt  jenkins_prod.txt
ubuntu@ip-172-31-31-182:~/jenkins/workspace/Job2$
```

i-02415f9acc7339d24 (prod)

PublicIPs: 50.17.67.252 PrivateIPs: 172.31.31.182

----------END ----------

Assignment 3 =>

Tasks To Be Performed:

1. Create a pipeline in Jenkins

2. Once push is made to “develop” a branch in Git, trigger job “test”. This will copy Git files to test node

3. If test job is successful, then prod job should be triggered

4. Prod jobs should copy files to prod node

setting up pipeline =>

Not secure 54.90.174.91:8080/job/CI-CD%20Pipeline/configure

Jenkins / CI-CD Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Days to keep builds

if not empty, build records are only kept up to this number of days

Max # of builds to keep

if not empty, only up to this number of build records are kept

10

Advanced

- [x] Do not allow concurrent builds
  - [ ] Abort previous builds

Jenkins / CI-CD Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- [ ] Build after other projects are built
- [ ] Build periodically
- [x] GitHub hook trigger for GITScm polling
- [ ] Poll SCM
- [ ] Trigger builds remotely (e.g., from scripts)

Jenkins / CI-CD Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script from SCM

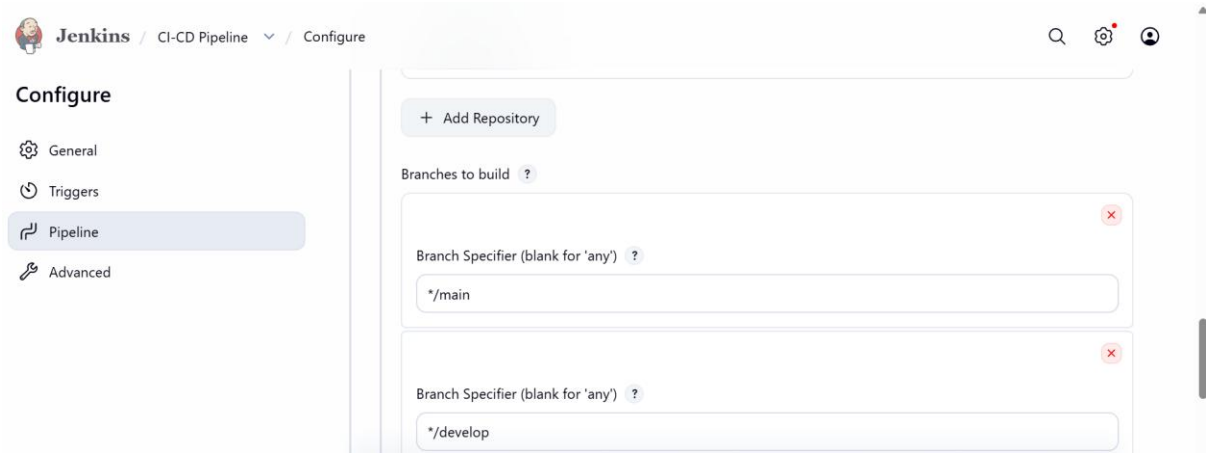
SCM

Git

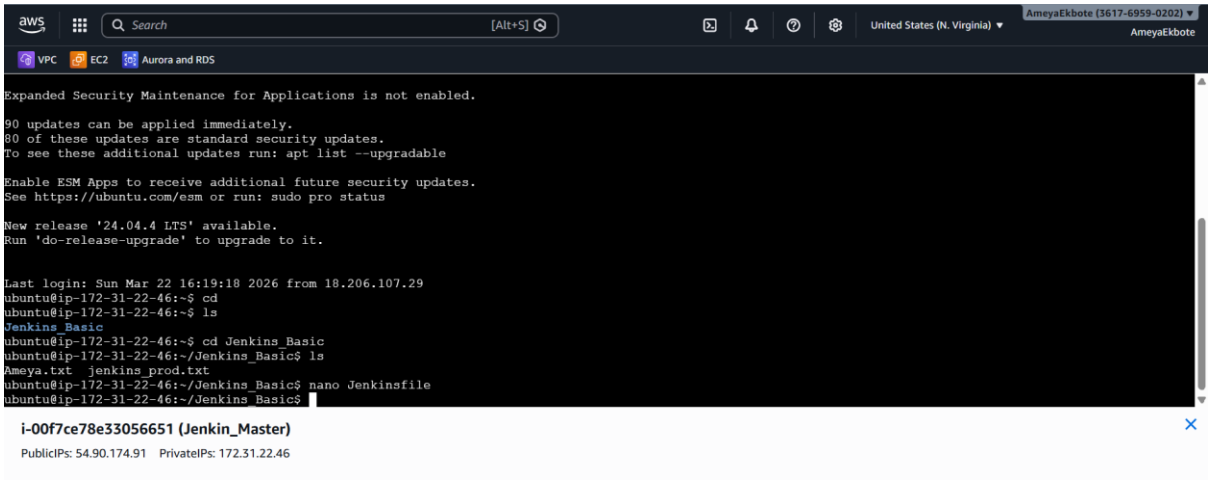
Repositories

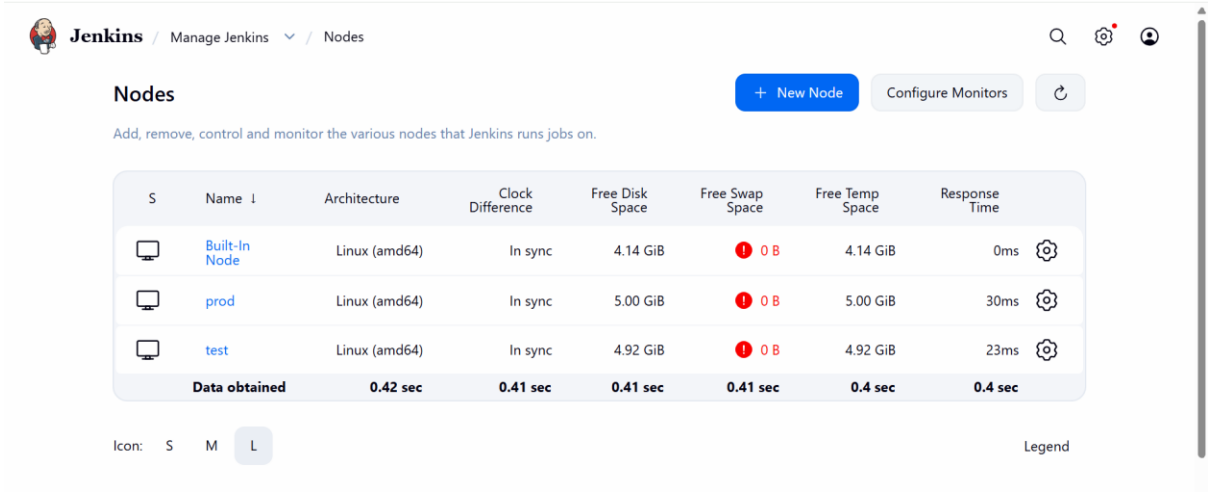
Repository URL

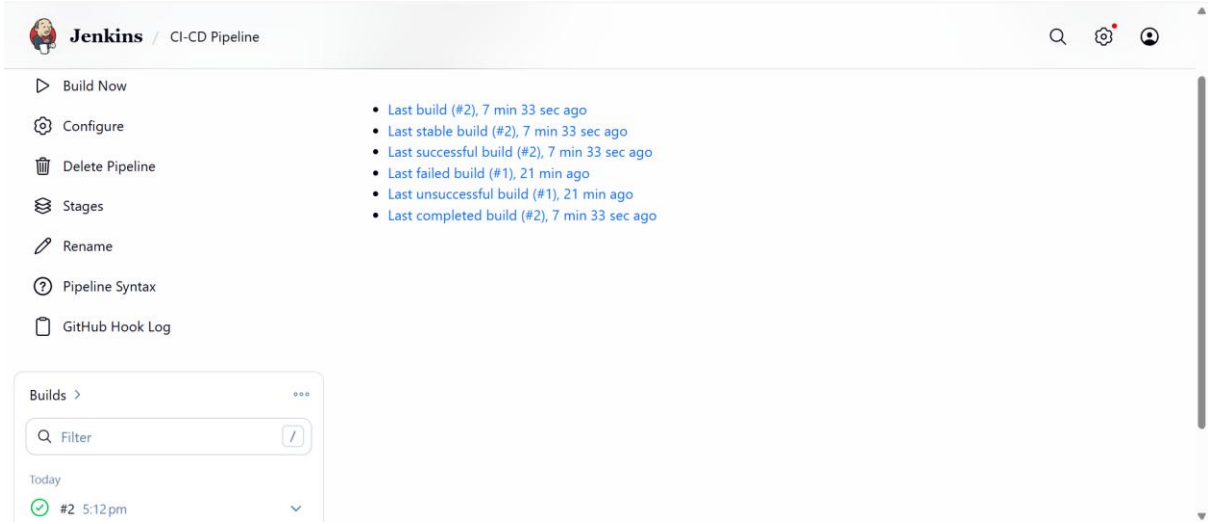
https://github.com/Ameya75/Jenkins_Basic.git




```
Jenkinfiles =>

pipeline {
    agent none
    stages {
        stage('Test Deploy') {
            agent { label 'test' }
            when {
                expression { env.GIT_BRANCH == 'origin/develop' }
            }
            steps {
                git branch: 'develop',
                    url: 'https://github.com/Ameya75/Jenkins_Basic.git'
                sh 'mkdir -p /home/ubuntu/test-deploy'
                sh 'cp -r * /home/ubuntu/test-deploy/'
            }
        }
        stage('Prod Deploy') {
            agent { label 'prod' }
            when {
                expression { env.GIT_BRANCH == 'origin/main' }
            }
            steps {
                git branch: 'main',
                    url: 'https://github.com/Ameya75/Jenkins_Basic.git'
                sh 'mkdir -p /home/ubuntu/prod-deploy'
                sh 'cp -r * /home/ubuntu/prod-deploy/'
            }
        }
    }
}
```








```
git checkout develop
```

```
git push origin develop
```

```
ubuntu@ip-172-31-31-106:~$ ls
jenkins  test-deploy
ubuntu@ip-172-31-31-106:~$ cd test-deploy/
ubuntu@ip-172-31-31-106:~/test-deploy$ ls
Ameya.txt  Ameya_Dev.txt  Jenkinsfile  jenkins.txt
ubuntu@ip-172-31-31-106:~/test-deploy$
```

i-0ec84811adcc311bc (test)

PublicIPs: 52.90.100.113 PrivateIPs: 172.31.31.106

```
ubuntu@ip-172-31-31-182:~$ ls
jenkins
ubuntu@ip-172-31-31-182:~$ ls
jenkins
ubuntu@ip-172-31-31-182:~$ ls
jenkins
ubuntu@ip-172-31-31-182:~$ ls
jenkins  prod-deploy
ubuntu@ip-172-31-31-182:~$ cd prod-deploy/
ubuntu@ip-172-31-31-182:~/prod-deploy$ ls
Ameya.txt  Ameya_Dev.txt  Jenkinsfile  jenkins_prod.txt
ubuntu@ip-172-31-31-182:~/prod-deploy$
```

i-02415f9acc7339d24 (prod)

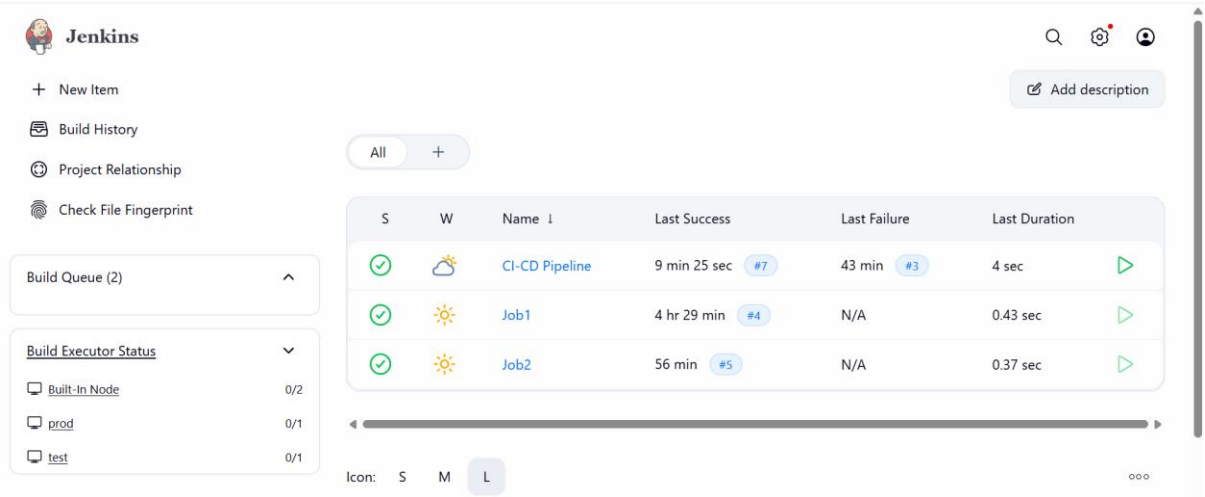
PublicIPs: 50.17.67.252 PrivateIPs: 172.31.31.182

```
Output from Pipeline =>

Started by GitHub push by Ameya75
Lightweight checkout support not available, falling back to full checkout.
Checking out git https://github.com/Ameya75/Jenkins_Basic.git into
/var/lib/jenkins/workspace/CI-CD
Pipeline@script/8431886c1d06be834d1fa6451a0b69b86432e7b88e23ac8981e1f72abcd
c1cd4 to read Jenkinsfile
The recommended git tool is: NONE
No credentials specified
 > git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/CI-CD
Pipeline@script/8431886c1d06be834d1fa6451a0b69b86432e7b88e23ac8981e1f72abcd
c1cd4/.git # timeout=10
Fetching changes from the remote Git repository
 > git config remote.origin.url
https://github.com/Ameya75/Jenkins_Basic.git # timeout=10
Fetching upstream changes from https://github.com/Ameya75/Jenkins_Basic.git
 > git --version # timeout=10
 > git --version # 'git version 2.34.1'
```

```
 > git fetch --tags --force --progress --
https://github.com/Ameya75/Jenkins_Basic.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
Seen branch in repository origin/develop
Seen branch in repository origin/main
Seen branch in repository origin/master
Seen 3 remote branches
 > git show-ref --tags -d # timeout=10
Checking out Revision c0894b19158da57983bff476ac65e40a3a85f6e2
(origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f c0894b19158da57983bff476ac65e40a3a85f6e2 # timeout=10
Commit message: "Recreate Ameya_Dev.txt and corrected Jenkinsfile"
 > git rev-list --no-walk 78aa83f678ef5ea25653273c5e1769a59481acc0 #
timeout=10
[Pipeline] Start of Pipeline
[Pipeline] stage
[Pipeline] { (Test Deploy)
[Pipeline] node
Running on test in /home/ubuntu/jenkins/workspace/CI-CD Pipeline
[Pipeline] {
[Pipeline] checkout
The recommended git tool is: NONE
No credentials specified
Fetching changes from the remote Git repository
Checking out Revision c0894b19158da57983bff476ac65e40a3a85f6e2
(origin/main)
Commit message: "Recreate Ameya_Dev.txt and corrected Jenkinsfile"
[Pipeline] withEnv
[Pipeline] {
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
Stage "Test Deploy" skipped due to when conditional
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Prod Deploy)
[Pipeline] node
Running on prod in /home/ubuntu/jenkins/workspace/CI-CD Pipeline
[Pipeline] {
[Pipeline] checkout
The recommended git tool is: NONE
No credentials specified
```

```
Fetching changes from the remote Git repository
Checking out Revision c0894b19158da57983bff476ac65e40a3a85f6e2
(origin/main)
Commit message: "Recreate Ameya_Dev.txt and corrected Jenkinsfile"
[Pipeline] withEnv
[Pipeline] {
 > git rev-parse --resolve-git-dir /home/ubuntu/jenkins/workspace/CI-CD
Pipeline/.git # timeout=10
 > git config remote.origin.url
https://github.com/Ameya75/Jenkins_Basic.git # timeout=10
Fetching upstream changes from https://github.com/Ameya75/Jenkins_Basic.git
 > git --version # timeout=10
 > git --version # 'git version 2.34.1'
 > git fetch --tags --force --progress --
https://github.com/Ameya75/Jenkins_Basic.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
Seen branch in repository origin/develop
Seen branch in repository origin/main
Seen branch in repository origin/master
Seen 3 remote branches
 > git show-ref --tags -d # timeout=10
 > git config core.sparsecheckout # timeout=10
 > git checkout -f c0894b19158da57983bff476ac65e40a3a85f6e2 # timeout=10
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Fetching changes from the remote Git repository
Checking out Revision c0894b19158da57983bff476ac65e40a3a85f6e2
(refs/remotes/origin/main)
Commit message: "Recreate Ameya_Dev.txt and corrected Jenkinsfile"
[Pipeline] sh
 > git rev-parse --resolve-git-dir /home/ubuntu/jenkins/workspace/CI-CD
Pipeline/.git # timeout=10
 > git config remote.origin.url
https://github.com/Ameya75/Jenkins_Basic.git # timeout=10
Fetching upstream changes from https://github.com/Ameya75/Jenkins_Basic.git
 > git --version # timeout=10
 > git --version # 'git version 2.34.1'
 > git fetch --tags --force --progress --
https://github.com/Ameya75/Jenkins_Basic.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
Seen branch in repository origin/develop
Seen branch in repository origin/main
Seen branch in repository origin/master
Seen 3 remote branches
 > git show-ref --tags -d # timeout=10
```

```
 > git config core.sparsecheckout # timeout=10
 > git checkout -f c0894b19158da57983bff476ac65e40a3a85f6e2 # timeout=10
+ mkdir -p /home/ubuntu/prod-deploy
[Pipeline] sh
 > git rev-parse --resolve-git-dir /home/ubuntu/jenkins/workspace/CI-CD
Pipeline/.git # timeout=10
 > git config remote.origin.url
https://github.com/Ameya75/Jenkins_Basic.git # timeout=10
Fetching upstream changes from https://github.com/Ameya75/Jenkins_Basic.git
 > git --version # timeout=10
 > git --version # 'git version 2.34.1'
 > git fetch --tags --force --progress --
https://github.com/Ameya75/Jenkins_Basic.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
 > git config core.sparsecheckout # timeout=10
 > git checkout -f c0894b19158da57983bff476ac65e40a3a85f6e2 # timeout=10
 > git branch -a -v --no-abbrev # timeout=10
 > git checkout -b main c0894b19158da57983bff476ac65e40a3a85f6e2 #
timeout=10
+ cp -r Ameya.txt Ameya_Dev.txt Jenkinsfile jenkins_prod.txt
/home/ubuntu/prod-deploy/
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] }
[Pipeline] // stage
[Pipeline] End of Pipeline
Finished: SUCCESS
```



----------END----------